

STensor/Hypersurface (Experimental)

Setup

```
In[*]:= << mGRG`STensor`
```

```
In[*]:= $PreRead = ReplaceAll[#,  
    expr_String => StringReplace[expr, "SP" -> "mGRG`STensor`Private"]] &;
```

```
In[*]:= DefKind[Greek, Alphabet["Greek"]]
```

Define Hypersurface

DefHypersurface

```
In[*]:= DefHypersurface::usage
```

```
Out[*]=
```

```
DefHypersurface[subKind, normSign] defines the geometric  
objects for a hypersurface. 'subKind' is a symbol for the  
kind on the hypersurface, and 'normSign' specifies the norm  
of the normal vector (-1 for spacelike, +1 for timelike).
```

```
In[*]:= SetDefaultKind[Greek]
```

```
In[*]:= DefHypersurface[Latin]
```

Predefined tensors including the metric and the covariant derivative:

```
In[*]:= SubBasis::usage
```

```
Out[*]=
```

```
SubBasis[a, b] is the projector tensor that maps  
vectors from the ambient spacetime to the hypersurface.
```

```
In[*]:= NormalV::usage
```

```
Out[*]=
```

```
NormalV[a] represents the normal vector to the hypersurface.
```

In[*]:= **ExtrinsicK::usage**

Out[*]=

ExtrinsicK[a, b] represents the extrinsic curvature tensor (or second fundamental form) of the hypersurface.

In[*]:= **SubMetric::usage**

Out[*]=

SubMetric[a, b] represents the induced metric (or first fundamental form) on the hypersurface.

In[*]:= **SubBasis[la, uα]**
NormalV[uα]
ExtrinsicK[la, lb]
SubMetric[la, lb]

Out[*]=

h_a^α

Out[*]=

n^α

Out[*]=

K_{ab}

Out[*]=

γ_{ab}

In[*]:= **SubCD::usage**

Out[*]=

SubCD[a, T] represents the intrinsic covariant derivative on the hypersurface, compatible with the induced metric SubMetric.

In[*]:= **SubCD[la, SubMetric[lb, lc]] (* metric compatible derivative *)**

Out[*]=

0

In[*]:=
GammaSubCD[la, lb, uc]
RicciSubCD[la, lb]
RiemannSubCD[la, lb, lc, ud]
ScalarSubCD[]

Out[*]=

$$\Gamma[D]_{ab}{}^c$$

Out[*]=

$$R[D]_{ab}$$

Out[*]=

$$R[D]_{abc}{}^d$$

Out[*]=

$$R[D]$$

In[*]:=
GetEpsilon[Greek]
GetEpsilon[Latin]

Out[*]=

Epsilon

Out[*]=

EpsilonLatin

In[*]:=
Epsilon[lμ, lv, lρ, lσ]
EpsilonLatin[la, lb, lc]

Out[*]=

$$\epsilon_{\mu\nu\rho\sigma}$$

Out[*]=

$$\epsilon[\text{Latin}]_{abc}$$

Rules:

In[*]:=	$\begin{aligned} &\text{NormalV}[\text{la}] \times \text{NormalV}[\text{ua}] \\ &\text{SubBasis}[\text{la}, \text{ua}] \times \text{NormalV}[\text{la}] \\ &\text{SubBasis}[\text{la}, \text{ua}] \times \text{SubBasis}[\text{lb}, \text{ub}] \times \text{Metricg}[\text{la}, \text{lb}] \\ &\text{SubBasis}[\text{la}, \text{ua}] \times \text{SubBasis}[\text{lb}, \text{la}] \end{aligned}$
Out[*]=	-1
Out[*]=	0
Out[*]=	γ_{ab}
Out[*]=	γ_{ab}

Tangential derivative $h_a^\alpha \nabla_\alpha$:

In[*]:=	TangentialD::usage
Out[*]=	TangentialD[a, expr] represents the tangential derivative of 'expr' along the direction of the index 'a' on the hypersurface.
In[*]:=	TangentialD[la, SubBasis[lb, ua]] (* $h_a^\beta \nabla_\beta h_b^\alpha = \Gamma_{ab}^c h_c^\alpha - \epsilon K_{ab} n^\alpha$ *)
Out[*]=	$K_{ab} n^\alpha + \Gamma[D]_{ab}^c h_c^\alpha$
In[*]:=	TangentialD[la, NormalV[ua]] (* $h_a^\beta \nabla_\beta n^\alpha = K_a^b h_b^\alpha$ *)
Out[*]=	$K_a^b h_b^\alpha$

Derivation of Gauss-Codazzi equations

$$\begin{aligned} h_c^\gamma \nabla_\gamma (h_b^\beta \nabla_\beta h_a^\alpha) &= h_c^\gamma \nabla_\gamma (\Gamma_{ba}^d h_d^\alpha - \epsilon K_{ba} n^\alpha) \\ R_{abc}^d &= -(D_a \Gamma_{bc}^d - D_b \Gamma_{ac}^d + \Gamma_{ap}^d \Gamma_{bc}^p - \Gamma_{bp}^d \Gamma_{ac}^p) \\ (\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) T^\delta &= -R_{\alpha\beta\gamma}^\delta T^\gamma \end{aligned}$$

In[*]:=	RHS = TangentialD[lc, TangentialD[lb, SubBasis[la, ua]]]
Out[*]=	$\partial_c K_{ba} n^\alpha + K_{cd} \Gamma[D]_{ba}^d n^\alpha + \partial_c \Gamma[D]_{ba}^d h_d^\alpha + K_{ba} K_c^d h_d^\alpha + \Gamma[D]_{ba}^d \Gamma[D]_{cd}^e h_e^\alpha$

In[*]:=

SubBasis[uf, lα] RHS

Out[*]=

$$\partial_c \Gamma[D]_{ba}{}^f + K_{ba} K_c{}^f + \Gamma[D]_{ba}{}^d \Gamma[D]_{cd}{}^f$$

In[*]:=

-2 AntisymmetrizeIndices[%, {lb, lc}]

Out[*]=

$$\partial_b \Gamma[D]_{ca}{}^f - \partial_c \Gamma[D]_{ba}{}^f + K_b{}^f K_{ca} - K_{ba} K_c{}^f + \Gamma[D]_{bd}{}^f \Gamma[D]_{ca}{}^d - \Gamma[D]_{ba}{}^d \Gamma[D]_{cd}{}^f$$

$$R_{\gamma\beta\delta\alpha} h_a{}^\gamma h_b{}^\beta h_c{}^\delta h_d{}^\alpha = R_{abcd} + K_{ad} K_{cb} - K_{ac} K_{bd}$$

In[*]:=

NormalV[lα] RHS

Out[*]=

$$-\partial_c K_{ba} - K_{cd} \Gamma[D]_{ba}{}^d$$

In[*]:=

-2 AntisymmetrizeIndices[%, {lb, lc}]

Out[*]=

$$-\partial_b K_{ca} + \partial_c K_{ba} + K_{cd} \Gamma[D]_{ba}{}^d - K_{bd} \Gamma[D]_{ca}{}^d$$

$$R_{\gamma\beta\delta\alpha} h_a{}^\gamma h_b{}^\beta h_c{}^\delta n^\alpha = D_a K_{bc} - D_b K_{ac}$$

Basis expansion

One can regard $(h_a{}^\alpha, n^\alpha)$ be a basis of the whole manifold.

DecomposeTensor: $V^\alpha = (\gamma^{ab} h_b{}^\beta V_\beta) h_a{}^\alpha + \epsilon (n_\beta V^\beta) n^\alpha$

A Relativistic Toolkit (E. Poisson)

Unit normal vector

$$n^\alpha n_\alpha = \epsilon = \begin{cases} -1 & \text{if } \Sigma \text{ is spacelike} \\ +1 & \text{if } \Sigma \text{ is timelike} \end{cases}$$

Induced metric

Let $x^\alpha = x^\alpha(y^a)$, where y^a are the coordinates of Σ . Then the vectors $h_a{}^\alpha = \frac{\partial x^\alpha}{\partial y^a}$ are tangent to Σ : $h_a{}^\alpha n_\alpha = 0$. Now, for displacements within Σ we have

$$ds_\Sigma^2 = g_{\alpha\beta} dx^\alpha dx^\beta = g_{\alpha\beta} (h_a{}^\alpha dy^a) (h_b{}^\beta dy^b) = \gamma_{ab} dy^a dy^b,$$

where $\gamma_{ab} = h_a{}^\alpha h_b{}^\beta g_{\alpha\beta}$ is the induced metric, or first fundamental form of the hypersurface. The complete relations for the inverse metric is given by

$$g^{\alpha\beta} = \epsilon n^\alpha n^\beta + \gamma^{ab} h_a{}^\alpha h_b{}^\beta.$$

Tangent tensor fields

Tensor fields $A^{\alpha\dots\beta}$ that are defined only on Σ and which are purely tangent to the hypersurface

admit the decomposition $A^{\alpha\dots\beta} = A^{a\dots b} h_a^\alpha \dots h_b^\beta$. In particular, we have

$$\gamma^{\alpha\beta} \equiv \gamma^{ab} h_a^\alpha h_b^\beta = g^{\alpha\beta} - \epsilon n^\alpha n^\beta.$$

The projections

$$A_{\alpha\dots\beta} h_a^\alpha \dots h_b^\beta = A_{a\dots b} \equiv \gamma_{am} \dots \gamma_{bn} A^{m\dots n}$$

give the tensor $A^{a\dots b}$ on Σ associated with $A^{\alpha\dots\beta}$.

Intrinsic covariant derivative

With a tangential vector A^α , we define the intrinsic covariant derivative $D_a A_b$ to be the projection of $\nabla_\alpha A_\beta$:

$$D_a A_b \equiv h_a^\alpha h_b^\beta \nabla_\alpha A_\beta = \partial_a A_b - \Gamma_{ab}^c A_c.$$

The connection Γ_{ab}^c is compatible with the induced metric γ_{ab} .

Extrinsic curvature

We decompose the vector $h_a^\alpha \nabla_\alpha A^\beta$ for a tangential vector A^α :

$$h_a^\alpha \nabla_\alpha A^\beta = (D_a A^\beta) h_b^\beta - \epsilon A^m (h_m^\mu h_a^\alpha \nabla_\alpha n_\mu) n^\beta.$$

With the definition of the extrinsic curvature, or second fundamental form $K_{ab} = h_a^\alpha h_b^\beta \nabla_\beta n_\alpha$, we have

$$h_b^\beta \nabla_\beta A^\alpha = (D_b A^\alpha) h_a^\alpha - \epsilon A^a K_{ab} n^\alpha, \quad (1)$$

that is, the tangential part of the vector field is $D_b A^a$, while the normal component is $-\epsilon A^a K_{ab}$. The extrinsic curvature is symmetric and

$$K_{ab} = h_a^\alpha h_b^\beta \nabla_{(\alpha} n_{\beta)} = \frac{1}{2} h_a^\alpha h_b^\beta \mathcal{L}_n g_{\alpha\beta}, \quad K = \gamma^{ab} K_{ab} = \nabla_\alpha n^\alpha.$$

Gauss-Codazzi equations

With $A^\alpha = h_b^\alpha$, $A^c = \delta_b^c$ and the equation (1) becomes

$$h_a^\beta \nabla_\beta h_b^\alpha = (D_a \delta_b^c) h_c^\alpha - \epsilon \delta_b^c K_{ca} n^\alpha = \Gamma_{ab}^c h_c^\alpha - \epsilon K_{ab} n^\alpha, \quad (2)$$

which is known as the Gauss-Weingarten equation. After some algebra we find

$$h_a^\alpha h_b^\beta h_c^\gamma R_{\alpha\beta\gamma}^\delta = R_{abc}^d h_d^\delta + \epsilon n^\delta (D_a K_{bc} - D_b K_{ac}) + \epsilon (K_{bc} h_a^\gamma - K_{ac} h_b^\gamma) \nabla_\gamma n^\delta.$$

Projecting along $h_{d\delta}$ gives

$$h_a^\alpha h_b^\beta h_c^\gamma h_d^\delta R_{\alpha\beta\gamma\delta} = R_{abcd} + \epsilon (K_{ad} K_{bc} - K_{ac} K_{bd}).$$

Projecting instead along n_δ gives

$$h_a^\alpha h_b^\beta h_c^\gamma n^\delta R_{\alpha\beta\gamma\delta} = D_a K_{bc} - D_b K_{ac}.$$

Generic decomposition

For a general vector V^α we have

$$V^\alpha = (\gamma^{ab} h_a^\beta h_b^\gamma V_\beta) h_b^\alpha + \frac{1}{\epsilon} (n^\beta V_\beta) n^\alpha$$

For $h_a^\beta \nabla_\beta n^\alpha$ we find

$$h_a^\beta \nabla_\beta n^\alpha = (V^{bc} h_b^\beta h_a^\gamma \nabla_\gamma n_\beta) h_c^\alpha + \frac{1}{\epsilon} (n^\beta h_a^\gamma \nabla_\gamma n_\beta) n^\alpha = K_a^b h_b^\alpha \quad (3)$$

A decomposition of a general tensor $T_{\alpha \dots \beta}{}^{\gamma \dots \delta}$ is equivalent to the basis expansion by (h_a^α, n^α) .

Therefore, for a general vector V^α we have the tangential derivative

$$\begin{aligned} h_a^\beta \nabla_\beta V^\alpha &= h_a^\beta \nabla_\beta (V^b h_b^\alpha + V^0 n^\alpha) = (h_a^\beta \nabla_\beta V^b) h_b^\alpha + V^b h_a^\beta \nabla_\beta h_b^\alpha + (h_a^\beta \nabla_\beta V^0) n^\alpha + V^0 (h_a^\beta \nabla_\beta n^\alpha) \\ &= (h_a^\beta \nabla_\beta V^b) h_b^\alpha + V^b (\Gamma_{ab}^c h_c^\alpha - \epsilon K_{ab} n^\alpha) + (h_a^\beta \nabla_\beta V^0) n^\alpha + V^0 (K_a^b h_b^\alpha) \\ &= (D_a V^b + K_a^b V^0) h_b^\alpha + (\partial_a V^0 - \epsilon K_{ab} V^b) n^\alpha \end{aligned}$$